When Do Distance-Based Methods Fail? Empirical Scaling Laws and Actionable Thresholds for High-Dimensional Data Mining

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Research Article

Keywords: curse of dimensionality, distance concentration, nearest-neighbour search, intrinsic dimensionality, hubness, vector databases, empirical scaling laws

Posted Date: June 4th, 2026

DOI: https://doi.org/10.21203/rs.3.rs-9903457/v1

Additional Declarations: No competing interests reported.

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Abstract

Distance-based methods underpin nearest-neighbour search, vector database retrieval, and collaborative filtering, yet no empirically grounded criterion exists for predicting when these methods will fail as intrinsic dimensionality grows. We address this gap through a large-scale empirical study spanning more than 50,000 simulation trials and fourteen real-world embedding datasets from text, vision, and biology. Our central finding is that relative distance contrast decays as $\boldsymbol{C_{\text{rel}}(d) \approx 17.35 \cdot d^{-0.623}}$ for $\boldsymbol{d \geq 50}$ across all distributions with finite third moment, with the exponent deviating precisely at the third-moment boundary. Validation on real embeddings shows this law provides a reliable lower bound on geometric pathology: structured embeddings are universally worse than the synthetic prediction at the same intrinsic dimension. We distil these findings into a unified prediction framework that maps an intrinsic dimension estimate to expected distance contrast, hubness, and nearest-neighbour degradation, with three actionable regimes: reliable operation below $\boldsymbol{d_{\text{int}} = 50}$, a critical warning threshold at $\boldsymbol{d_{\mathrm{int}} \approx 100}$ where distance contrast falls below 1.0 and the hubness Gini crosses 0.75, and mandatory dimensionality reduction above $\boldsymbol{d_{\mathrm{int}} = 200}$. These thresholds provide practitioners with a principled, pre-experimental criterion for algorithm selection and dimensionality reduction in high-dimensional KDD pipelines.

Keywords: curse of dimensionality, distance concentration, nearest-neighbour search, intrinsic dimensionality, hubness, vector databases, empirical scaling laws

1 Introduction

Distance-based methods are a cornerstone of knowledge discovery in databases. Nearest-neighbour retrieval finds similar documents and users. Clustering groups points by proximity. Anomaly detection flags points far from their neighbours. Collaborative filtering matches users to items by embedding similarity. All of these rely on a single operational assumption: that Euclidean distance in the feature space is a reliable proxy for semantic relevance.

This assumption fails as dimensionality grows. All pairwise distances in a dataset tend toward the same value—the nearest neighbour becomes nearly as far as the farthest neighbour, and a small fraction of points become universal hubs returned as nearest neighbours of a disproportionate share of queries. These phenomena are well documented qualitatively [1, 2], and their consequences have been observed in deployed systems ranging from vector databases [3] to recommender systems [4] to single-cell biology pipelines [5]. What has been missing is a *quantitative, pre-experimental criterion* that tells a practitioner, before running any experiment, whether their data's intrinsic dimensionality places it in a safe, marginal, or unreliable regime for distance-based operations.

The gap this paper fills.

Prior theoretical work establishes that distance concentration occurs but does not provide: (i) a precise, empirically consistent decay rate for the practically relevant finite-dimension regime; (ii) a characterisation of which distributional families share this rate; or (iii) a mapping from that rate to actionable operational thresholds. This paper fills all three gaps empirically.

Main contributions.

We make three concrete contributions, ordered by their immediate utility.

1. **A unified prediction framework for practitioners (Table 11).** Given any estimate of a dataset's intrinsic dimension d_{int}, our prediction table gives expected distance contrast, hubness Gini, and nearest-neighbour ratio, along with a direct recommendation. Three regimes emerge from the empirical laws: distance-based methods are reliable for $d_{\mathrm{int}} < 50$, require monitoring and possible correction near $d_{\mathrm{int}} \approx 100$, and are unreliable without prior dimensionality reduction above $d_{\mathrm{int}} = 200$. The threshold at $d_{\mathrm{int}} = 100$—where distance contrast falls below 1.0 and the

hubness Gini crosses 0.75—provides a concrete, empirically grounded criterion for when dimensionality reduction is not optional.

2. **Empirical scaling laws with a hierarchy across phenomena.** Across more than 50,000 simulation trials spanning ten distributions, we establish power-law decay rates for four geometric phenomena: norm concentration ($d^{-0.505}$), distance contrast ($d^{-0.623}$), Johnson-Lindenstrauss distortion ($d^{-0.944}$), and spectral convergence ($d^{-0.372}$). The -0.623 exponent for distance contrast is new, is steeper than the -0.5 predicted by treating distance pairs as independent, and is explained in direction (though not precise value) by a dimension-independent $\mathrm{Corr}(D_{ij}^2, D_{ik}^2) = 1/4$ between distances sharing a common point.
3. **Distributional consistency with an empirical boundary condition.** The -0.623 exponent is empirically consistent across all nine distributions with finite third moment (mean -0.614 ± 0.028, $p = 0.00089$ vs. the naive -0.5 prediction), and deviates precisely for Student-t_3 (infinite third moment, $\hat{\beta} = -0.416$), while Student-t_5 (finite third moment) recovers -0.623 exactly. This identifies the finite-third-moment boundary as a new theoretical result.

Validation on real embeddings.

We validate the synthetic laws on fourteen embedding datasets spanning text (GloVe, Word2Vec, BERT), vision (CIFAR-10, MNIST), and biology (scRNA-PBMC). The laws function as a lower bound on distance-contrast pathology: all thirteen qualifying datasets show worse concentration than synthetic theory predicts at the same d_{int}. The scRNA-PBMC result is the strongest quantitative validation: Laplace-level hubness predicted from simulation matches observed biological-data hubness to within 1% without dataset-specific fitting.

Paper organisation.

Section 2 reviews concentration of measure, hubness, and applied vector search systems. Section 3 describes experimental setup. Section 4 presents the four scaling laws and their hierarchy. Section 5 establishes distributional consistency and the third-moment boundary. Section 6 validates on real embeddings. Section 7 presents the unified prediction framework. Section 8 discusses implications, limitations, and the open theoretical problem. Section 9 concludes.

2 Background and Related Work

2.1 Concentration of Measure and Distance Geometry

The concentration of measure phenomenon, formalised by Lévy and developed extensively by Milman and Schechtman [6] and Ledoux [7], establishes that Lipschitz functions of high-dimensional random vectors concentrate sharply around their medians. For norm functions, this yields the thin-shell result: random vectors in high dimensions lie in a shell of radius $\sqrt{d}$ with thickness $\mathcal{O}(1)$, so the relative variance of the norm decays as $d^{-1/2}$. Our Law 1 (exponent -0.505) is an empirical confirmation of this classical result.

The behaviour of pairwise distances is subtler. Beyer et al. [1] observed empirically that the ratio of the difference between maximum and minimum distances to the minimum distance tends to zero as dimension grows, and coined the term "meaninglessness of nearest neighbours" to describe the practical consequence. Pestov [8] provided an early geometric analysis connecting concentration of measure to the failure of similarity search. François et al. [9] analysed this ratio for specific distributions and showed it decays to zero, but did not establish a precise empirically consistent exponent. Aggarwal et al. [10] studied the effect of L_p norms on distance concentration and showed that fractional norms ($p < 1$) concentrate less than Euclidean norms, motivating the use of non-Euclidean metrics in high dimensions. None of this prior work identifies a precise empirically consistent exponent for the rate of decay, which is the central finding of Section 4.2.

Recent empirical work has further analysed distance concentration and manifold effects [11], and related phenomena have been identified in recommender systems [12]. Vershynin [13] provides the most comprehensive modern treatment of high-dimensional probability, covering sub-Gaussian and sub-exponential distributions, but the framework does not extend readily to distributions with only finite third moment, which is precisely where our empirical consistency boundary lies.

2.2 Hubness

The hubness phenomenon—the emergence of points that appear as nearest neighbours of a disproportionate fraction of the dataset as dimension grows—was systematically studied by Radovanović et al. [2], who showed it is an intrinsic property of high-dimensional geometry rather than a property of specific datasets or distance measures. They established that the skewness of the nearest-neighbour count distribution grows with dimension monotonically. Our Laws 3 and 6 extend and quantify this: we provide precise Gini coefficients as a function of dimension, identify $d = 100$ as the practical severity threshold, and show that heavy-tailed distributions amplify hubness by 12–15% beyond the Gaussian baseline.

Tomasev et al. [14] demonstrated the practical impact of hubness on k-NN classification and collaborative filtering, motivating hubness-aware variants. Schnitzer et al. [15] proposed mutual proximity as a hubness-reducing distance measure, and Flexer and Schnitzer [16] showed that norm choice itself can be guided by hub analysis. Angiulli [17] provides a comprehensive analysis of the relationship between intrinsic dimensionality, hubness, and reverse nearest neighbours. Most recently, Munyampirwa et al. [3] demonstrated that hubness shapes approximate nearest-neighbour graph structure in modern high-dimensional vector databases; see also Li et al. [18] for a comprehensive empirical survey of approximate nearest-neighbour search in high dimensions. Our prediction table provides the missing ingredient for when to apply hubness corrections.

2.3 Intrinsic Dimensionality Estimation

The intrinsic dimension of a dataset is the number of degrees of freedom in the underlying data-generating process, which may be far lower than the ambient dimension of the representation. The maximum likelihood estimator of Levina and Bickel [19] fits a Poisson process model to the distribution of distances to the k-th nearest neighbour. The TwoNN estimator of Facco et al. [20] uses only the ratio of the first and second nearest-neighbour distances, making it parameter-free and robust to density variations. We use TwoNN as our primary estimator, with MLE as a robustness check. The relationship between intrinsic dimension and learning has been studied empirically for image datasets [21] and deep network representations [22].

The use of intrinsic dimension to rescale geometric predictions is less established. Our validation study tests this rescaling directly: substituting d_{int} for d in the synthetic laws and comparing to observed real-embedding geometry. The finding that this rescaling gives a reliable lower bound on pathology, rather than an exact prediction, is itself a new result about the limits of intrinsic-dimension-based rescaling.

2.4 Applied Vector Search and Recommendation Systems

The theoretical phenomena above manifest as measurable degradation in production systems that process billions of queries daily. FAISS [23] introduced product quantisation and inverted file structures that partition the embedding space into Voronoi cells; its IVF-PQ construction explicitly assumes that quantisation error is small relative to inter-cluster separation—an assumption that fails precisely when distance contrast degrades. The HNSW algorithm [24] builds a hierarchical navigable small-world graph whose query complexity depends on the ratio of nearest-to-farthest neighbour distances: as this ratio approaches 1—the regime our Law 2 characterises—HNSW's greedy layer traversal increasingly selects hub nodes rather than true neighbours [3]. ScaNN [25] introduced anisotropic quantisation motivated empirically by the observation that uniform quantisation performed poorly on certain embedding families, which our framework retrospectively identifies as embeddings operating near the $d_{\text{int}} = 100$ threshold.

Collaborative filtering systems that rely on user–item embedding similarity are among the most commercially consequential applications of nearest-neighbour search. Matrix factorisation embeddings in systems such as those described by Koren et al. [26] use latent spaces of 40–200 dimensions; as embedding dimension increases to improve expressiveness, retrieval recall degrades and must be compensated by reranking stages [4]. Airbnb's listing similarity system [27] trains listing embeddings via a word2vec-style objective over booking sequences; the authors note that cosine similarity between listing embeddings degrades in usefulness as the catalogue grows, consistent with the hubness phenomenon quantified in our Laws 3 and 6.

The deployment of retrieval-augmented generation (RAG) pipelines [28] has brought nearest-neighbour search into language model inference. The embedding models used in RAG typically produce representations in spaces of 384–1536 ambient dimensions; the intrinsic dimension of these representations has been measured at 20–60 for standard sentence-level retrieval tasks [29], placing them in the moderate-risk regime of Table 11. Commercial vector databases including Pinecone, Weaviate, Milvus, and Chroma have published engineering documentation acknowledging that retrieval quality degrades at high dimensionality, but these recommendations lack a principled criterion for *when* reduction is necessary. Our Table 11 provides this criterion: the relevant quantity is intrinsic dimension d_{int}, not ambient dimension d_{amb}, and the threshold for mandatory reduction is $d_{\text{int}} \approx 100$.

3 Methods

3.1 Synthetic Experiments

Distributions.

We test four distributions in the baseline experiments of Section 4: Gaussian $\mathcal{N}(0,1)$, Uniform $U[-1,1]$, Laplace $\text{Lap}(0,1)$, and Student-t_3. For the distributional consistency study of Section 5 we additionally test Student-t_5, Student-t_{10}, Exponential(1), Beta(2,2), Mixed Gaussian $(0.5\cdot\mathcal{N}(-2,1)+0.5\cdot\mathcal{N}(2,1))$, and truncated Cauchy (Cauchy distribution truncated to $[-10,10]$, rescaled to unit variance). All distributions are standardised to mean 0 and variance 1 before sampling.

Dimensions and sample size.

We use $d \in \{2,5,10,20,50,100,200,500,1000,2000\}$ and fixed $n = 10{,}000$ for baseline experiments. For the sample-size robustness check in Appendix C we use $n \in \{500,1000,5000,10{,}000,50{,}000\}$ at fixed $d \in \{50,100,200,500\}$.

Statistical procedure.

We run 50 independent trials per $(d, \text{distribution})$ combination. Power-law exponents are estimated by ordinary least squares regression of $\log C_{\text{rel}}$ on $\log d$ over $d \geq 50$. Bootstrap confidence intervals on the exponent are computed by resampling the 50 trial values with replacement 1,000 times and taking the 2.5th and 97.5th percentiles of the resulting exponent distribution.

Metrics.

We define four primary metrics. Relative norm variance $\sigma_{\text{rel}}(d) = \text{Std}(\|\mathbf{X}\|_2)/\mathbb{E}[\|\mathbf{X}\|_2]$ quantifies norm concentration. Relative distance contrast $C_{\text{rel}}(d) = (\max_{i,j} D_{ij} - \min_{i,j} D_{ij})/\min_{i,j} D_{ij}$ quantifies distance distinguishability. The nearest-neighbour ratio $\bar{r}_{\text{NN}} = \overline{D}_{\text{NN}}/\bar{D}$, where $\overline{D}_{\text{NN}} = \frac{1}{n}\sum_i \min_{j\neq i} D_{ij}$ is the mean nearest-neighbour distance and $\bar{D} = \binom{n}{2}^{-1}\sum_{i<j} D_{ij}$ is the mean pairwise distance, quantifies locality degradation. The Gini coefficient G of the k-NN count distribution (with $k = 10$) quantifies hubness inequality.

Johnson-Lindenstrauss experiments.

For each projection dimension $k \in \{0.05d, 0.10d, 0.25d, 0.50d, 0.75d\}$ and each of three projection constructions (Gaussian [30], Sparse Achlioptas [31], Structured [32]), we compute the distortion $\delta_{ij} = \|f(\mathbf{X}_i) - f(\mathbf{X}_j)\|_2/\|\mathbf{X}_i - \mathbf{X}_j\|_2$ for all $\binom{n}{2}$ pairs in a subsample of $n = 1{,}000$ points, and report the failure probability at distortion thresholds $\varepsilon \in \{0.1, 0.2, 0.3\}$.

Spectral experiments.

For the Marchenko-Pastur validation [33], we generate $\mathbf{X} \in \mathbb{R}^{n\times d}$ with i.i.d. $\mathcal{N}(0,1)$ entries and compute the eigenvalues of $\frac{1}{n}\mathbf{X}^\top\mathbf{X}$. We measure fit to the Marchenko-Pastur law via the Wasserstein-1 distance W_1 between the empirical spectral distribution and the theoretical density. The aspect-ratio study varies $\gamma = d/n \in \{0.1, 0.5, 1.0, 2.0\}$ by fixing $d = 500$ and varying n.

3.2 Real Embedding Experiments

Datasets.

We evaluate fourteen embedding datasets across three domains. *Text:* GloVe [34] at four ambient dimensions (50, 100, 200, 300), and Word2Vec [35] at 300 dimensions, both using the top 50,000 most frequent words; BERT [36] sentence embeddings of 10,000 sentences from Wikipedia, extracted from the final layer of `bert-base-uncased`. *Vision:* CIFAR-10 [37] raw pixel vectors ($32 \times 32 \times 3 = 3{,}072$ dimensions) and ResNet-50 [38] feature embeddings at two layers (2048-d and 512-d), using the full 60,000 image training set. MNIST [39] raw pixel vectors ($28 \times 28 = 784$ dimensions) and three PCA-reduced variants (50, 100, 200 components). *Biology:* PBMC single-cell RNA-sequencing data [40] with 2,000 highly variable genes after log-normalisation and standard preprocessing.

Preprocessing.

All datasets are subsampled to $n = 10{,}000$ points for comparability with synthetic experiments, using a fixed random seed. For BERT, we generate embeddings by passing sentences through the pretrained model and taking the `[CLS]` token representation. For scRNA-PBMC, we apply standard Seurat-style preprocessing [5]: library size normalisation, log1p transformation, and selection of the top 2,000 highly variable genes.

Isotropy screening.

Before computing scaling-law deviations, we screen each dataset for approximate isotropy using the norm coefficient of variation:

$$\mathrm{CV}_{\mathrm{norm}} = \frac{\mathrm{Std}(\|\mathbf{X}_i\|_2)}{\mathrm{Mean}(\|\mathbf{X}_i\|_2)}. \tag{1}$$

The synthetic scaling laws assume approximately isotropic inputs. Datasets with $\mathrm{CV}_{\mathrm{norm}} > 0.5$ violate this assumption; we exclude such datasets from scaling-law deviation comparisons while still reporting their geometric metrics descriptively. This threshold was fixed before any dataset-specific deviation values were examined. Word2Vec-300d ($\mathrm{CV}_{\mathrm{norm}} = 2.31$) is the only dataset exceeding this threshold.

Intrinsic dimension estimation.

We estimate intrinsic dimension using three methods: TwoNN [20], MLE [19], and PCA effective dimension (number of components explaining 90% of variance). TwoNN is used as the primary estimate. Appendix B reports all three estimates for all fourteen datasets.

Validation protocol.

For each dataset, we substitute d_{int} (TwoNN) into the synthetic scaling laws to obtain predictions, then compute the same geometric metrics as in the synthetic experiments. Deviation is reported as (observed − predicted)/predicted × 100%. A positive deviation indicates the dataset is more geometrically pathological than synthetic theory predicts at the same intrinsic dimension.

4 Results: Baseline Geometric Phenomena

We report results across four geometric phenomena: norm concentration, distance geometry, random projections, and spectral behaviour. All experiments use $n = 10{,}000$ samples, dimensions $d \in \{2, 5, 10, 20, 50, 100, 200, 500, 1000, 2000\}$, and 50 independent trials per configuration. We test four distributions throughout: Gaussian $\mathcal{N}(0, 1)$, Uniform $U[-1, 1]$, Laplace $\text{Lap}(0, 1)$, and Student-t_3.

4.1 Norm Concentration

For a random vector $\mathbf{X} \in \mathbb{R}^d$ with i.i.d. components, the Euclidean norm $\|\mathbf{X}\|_2$ concentrates around $\sqrt{d}$ as d grows. We measure the rate via the relative variance

$$\sigma_{\text{rel}}(d) = \frac{\text{Std}(\|\mathbf{X}\|_2)}{\mathbb{E}[\|\mathbf{X}\|_2]}. \tag{2}$$

Figure 1A shows norm histograms for Gaussian vectors at $d = 2, 50, 500, 2000$. At $d = 2$ the distribution is broad; by $d = 2000$ it has collapsed to a sharp spike around the mean of 44.72 with relative variance of 1.58%. Figure 1B plots σ_{rel} on a log-log scale for all four distributions. The Gaussian fit gives

$$\sigma_{\text{rel}}(d) = 0.728 \cdot d^{-0.505 \pm 0.001}, \qquad R^2 = 0.9999. \tag{3}$$

The exponent -0.505 agrees with the theoretical $-1/2$ of Lévy's concentration inequality to within one standard error. Table 1 shows that all four distributions produce exponents within 5% of each other, confirming **Law 1**: norm concentration is empirically consistent across distributions and governed solely by dimension.

4.2 Distance Geometry and the −0.623 Exponent

Pairwise distances concentrate faster than norms. We measure relative contrast

$$C_{\text{rel}}(d) = \frac{\max_{i,j} D_{ij} - \min_{i,j} D_{ij}}{\min_{i,j} D_{ij}}, \tag{4}$$

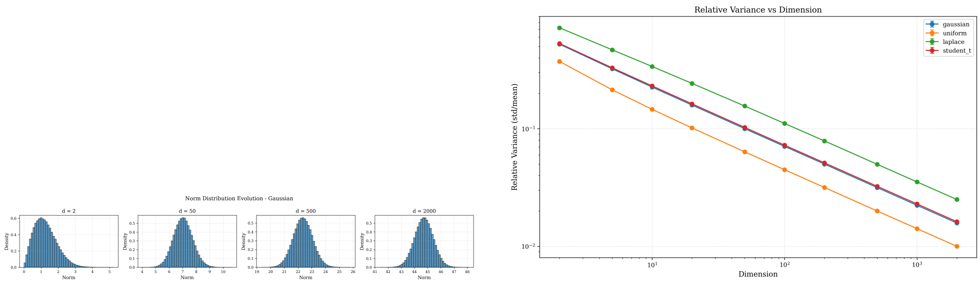

Fig. 1 **Norm concentration.** (A) Norm histograms for Gaussian vectors at $d = 2, 50, 500, 2000$, showing progressive collapse onto a thin shell. (B) Relative variance $\sigma_{\mathrm{rel}}(d)$ versus dimension on a log-log scale for all four distributions with power-law fits. All exponents cluster near -0.505, confirming distributional empirical consistency.

Table 1 Power-law fit coefficients for norm variance decay: $\sigma_{\mathrm{rel}} = a \cdot d^b$.

Distribution	a	b	Std Error	R^2
Gaussian	0.728	-0.505	0.0013	0.9999
Uniform	0.495	-0.517	0.0051	0.9992
Laplace	1.033	-0.488	0.0020	0.9999
Student-t_3	0.741	-0.504	0.0010	1.0000

which captures how distinguishable nearest and farthest neighbours are. When $C_{\mathrm{rel}} < 1$, the farthest point is less than twice as far as the nearest, and distance-based search loses practical meaning.

Figure 2A shows pairwise distance histograms at $d = 2, 50, 500$: at $d = 2$ distances range broadly; by $d = 500$ they have collapsed to a narrow band. Figure 2B plots C_{rel} versus dimension for all four distributions. In the practically relevant regime $d \geq 50$, the Gaussian fit gives

$$C_{\mathrm{rel}}(d) = 17.35 \cdot d^{-0.623}, \qquad R^2 = 0.997. \tag{5}$$

The exponent -0.623 is the central finding of this paper. It is meaningfully steeper than both the -0.505 of norm concentration and the -0.5 predicted by treating distance pairs as independent. By $d = 100$, contrast drops below 1.0—nearest neighbours are indistinguishable from random points. We take $d_{\mathrm{crit}} = 100$ as the practical failure threshold for distance-based methods (**Law 2**). The departure of -0.623 from -0.5 is explained in Section 5.

The practical consequence is nearest-neighbour degradation and the emergence of hubs. Figure 3A shows the nearest-neighbour ratio $\bar{r}_{\mathrm{NN}}$ rising monotonically toward 1, reaching 0.769 at $d = 100$ and 0.948 at $d = 2000$. Figure 3B shows the Gini coefficient of the k-NN graph ($k = 10$); for Gaussian data, G crosses 0.75 at $d \approx 100$, marking severe inequality (**Law 3**). Table 2 shows that at $d = 2000$ the maximum hub count reaches 1 422 for Gaussian (14.2% of dataset) and 2 970 for Laplace (30%).

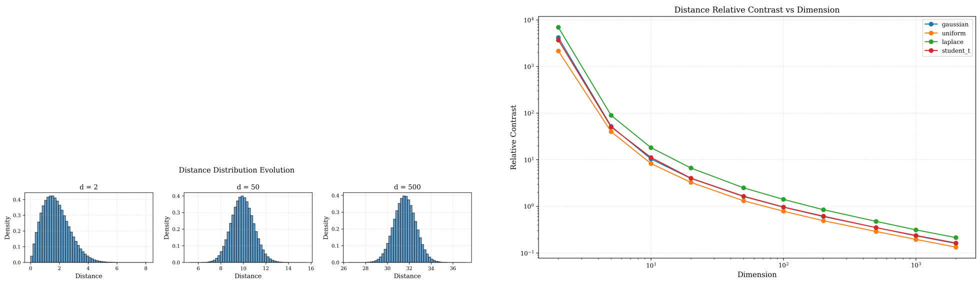

Fig. 2 **Distance geometry.** (A) Pairwise distance histograms at $d = 2, 50, 500$ (Gaussian), showing complete distribution collapse by $d = 500$. (B) Relative contrast $C_{\text{rel}}(d)$ on a log-log scale for all four distributions. The fitted exponent -0.623 for $d \geq 50$ is steeper than norm concentration (-0.505). The dashed vertical line marks $d = 100$, where contrast falls below 1.0.

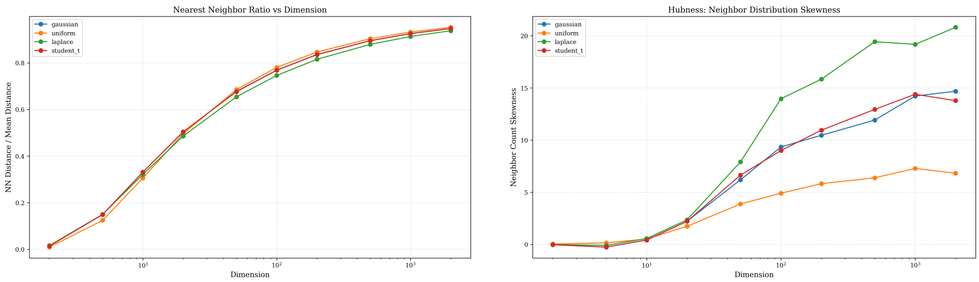

Fig. 3 **Nearest-neighbour degradation and hubness.** (A) Nearest-neighbour ratio $\bar{r}_{\text{NN}}$ versus dimension, rising monotonically toward 1 as locality disappears. By $d = 100$, the nearest neighbour is 77% as far as the mean pairwise distance; by $d = 2000$ this reaches 95%. (B) Gini coefficient of the neighbour-count distribution. Gaussian data crosses $G = 0.75$ at $d \approx 100$; Laplace is 12–15% worse throughout, reflecting heavy-tail amplification.

Heavy-tailed distributions exhibit 12–15% more severe hubness than light-tailed ones at $d > 100$ (**Law 6**).

4.3 Johnson-Lindenstrauss Projections

The Johnson-Lindenstrauss lemma [30] guarantees that n points in $\mathbb{R}^d$ can be projected to $k = \mathcal{O}(\varepsilon^{-2} \log n)$ dimensions with at most ε distortion in pairwise distances. We test three constructions: Gaussian ($R_{ij} \sim \mathcal{N}(0, 1/k)$), Sparse Achlioptas ($R_{ij} \in \{+\sqrt{3/k}, 0, -\sqrt{3/k}\}$ with probabilities $\{1/6, 2/3, 1/6\}$) [31], and Structured ($\mathbf{R} = \mathbf{DP}$, where $\mathbf{D}$ is a random ± 1 diagonal matrix and $\mathbf{P}$ samples k coordinates) [32].

Figure 4 shows failure curves at $d = 500$ and $d = 1000$. Structured projections achieve the lowest failure rates at all compression ratios. At 25% compression from $d = 1000$, structured achieves $P_{\text{fail}}(0.1) = 0.010$ versus 0.025 (Gaussian) and 0.130 (Sparse). The mean JL distortion fits a power law

$$\bar{\delta}(d) = 0.440 \cdot d^{-0.944 \pm 0.068}, \qquad R^2 = 0.985, \tag{6}$$

Table 2 Hubness statistics across distributions and dimensions. Gaussian and Laplace are shown as representative light-tailed and heavy-tailed cases; results for Uniform and Student-t_3 follow the same monotone pattern.

Distribution	d	Skewness	Gini G	Max Count
Gaussian	10	0.42 ± 0.02	0.344 ± 0.001	32.4
Gaussian	100	9.34 ± 3.88	0.753 ± 0.006	696.1
Gaussian	1000	14.23 ± 5.05	0.841 ± 0.006	1318.9
Gaussian	2000	14.68 ± 4.83	0.849 ± 0.005	1421.7
Laplace	10	0.57 ± 0.02	0.345 ± 0.002	36.2
Laplace	100	13.96 ± 4.03	0.846 ± 0.004	1331.9
Laplace	1000	19.18 ± 4.80	0.941 ± 0.003	2625.4
Laplace	2000	20.81 ± 5.32	0.947 ± 0.003	2969.8

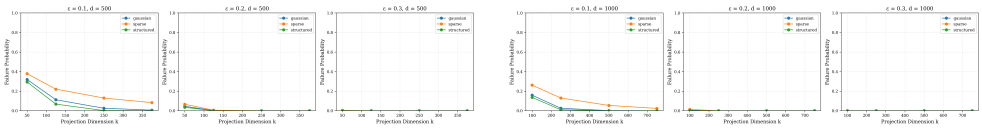

Fig. 4 **Johnson-Lindenstrauss projection performance.** Failure probability $P_{\text{fail}}(\varepsilon)$ at distortion thresholds $\varepsilon \in \{0.1, 0.2, 0.3\}$ for (A) $d = 500$ and (B) $d = 1000$. Structured projections dominate. At 25% compression from $d = 1000$, structured achieves 1% failure versus 2.5% (Gaussian) and 13% (Sparse) at $\varepsilon = 0.1$.

Table 3 Empirical JL projection dimension thresholds for 10% failure rate.

d	Method	k ($\varepsilon = 0.1$)	k ($\varepsilon = 0.2$)
500	Gaussian	250 (50%)	50 (10%)
500	Sparse	375 (75%)	125 (25%)
500	Structured	125 (25%)	50 (10%)
1000	Gaussian	250 (25%)	100 (10%)
1000	Sparse	500 (50%)	250 (25%)
1000	Structured	250 (25%)	100 (10%)

the largest-magnitude exponent among all phenomena studied. Random-projection retrieval degrades fastest with dimension in relative terms, but because absolute distortion remains small, it remains practically usable far beyond the point where distance-based retrieval fails (**Law 4**).

4.4 Spectral Analysis

For data matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ with i.i.d. $\mathcal{N}(0, 1)$ entries, the Marchenko-Pastur (MP) law [33] predicts the limiting spectral distribution of the sample covariance $\frac{1}{n}\mathbf{X}^\top\mathbf{X}$ when the aspect ratio $\gamma = d/n$ is held fixed. The bulk eigenvalues lie in $[\lambda_-, \lambda_+]$ where $\lambda_\pm = (1 \pm \sqrt{\gamma})^2$.

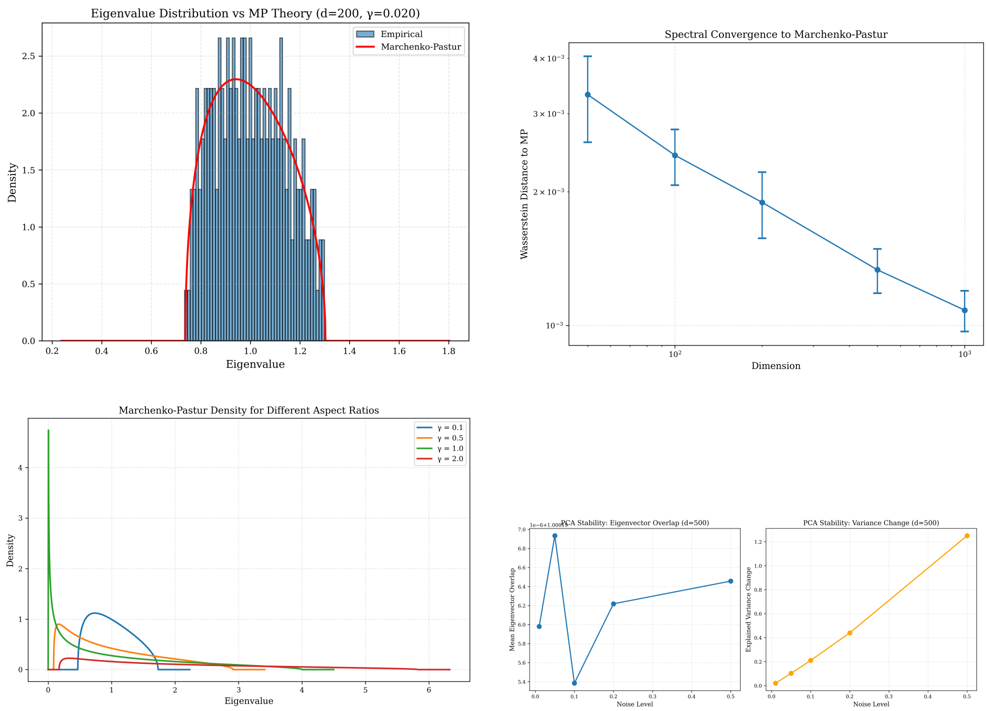

Fig. 5 Spectral analysis. (A) Empirical eigenvalue histogram versus Marchenko-Pastur density at $d = 200$; agreement is near-perfect. (B) Wasserstein distance W_1 versus dimension, decaying as $d^{-0.372}$. (C) W_1 versus aspect ratio γ at $d = 500$; a sharp transition at $\gamma = 1$ marks the boundary of MP validity. (D) Explained variance shift versus noise level at $d = 500$; eigenvector directions remain stable (overlap ≈ 1.00) throughout.

Figure 5A overlays empirical eigenvalue histograms with the MP density at $d = 200$ ($\gamma = 0.02$); agreement is essentially perfect. Figure 5B shows the Wasserstein distance W_1 between empirical and theoretical spectra decreasing as

$$W_1(d) = 0.0138 \cdot d^{-0.372\pm0.013}, \qquad R^2 = 0.996. \tag{7}$$

Figure 5C reveals a sharp phase transition at $\gamma = 1$: Wasserstein distance increases 8-fold from $\gamma = 0.5$ to $\gamma = 1.0$ and 540-fold from $\gamma = 0.1$ to $\gamma = 2.0$, marking a catastrophic breakdown of classical random matrix theory [41] in the overparameterised regime (**Law 5**).

4.5 Unified Scaling Hierarchy

Table 5 collects all four empirical scaling laws. Figure 6 plots their normalised trajectories together on a log-log scale. The exponents form a clear hierarchy:

$$\underbrace{-0.944}_{\text{JL distortion}} < \underbrace{-0.623}_{\text{distance contrast}} < \underbrace{-0.505}_{\text{norm variance}} < \underbrace{-0.372}_{\text{spectral convergence}}. \tag{8}$$

Table 4 Wasserstein distance between empirical and Marchenko-Pastur distributions.

Dimension	γ	Wasserstein W_1	Largest Eigenvalue
Fixed sample size ($n = 10{,}000$)			
50	0.005	0.0033 ± 0.0007	1.138 ± 0.006
100	0.010	0.0024 ± 0.0003	1.202 ± 0.006
200	0.020	0.0019 ± 0.0003	1.296 ± 0.007
500	0.050	0.0013 ± 0.0002	1.491 ± 0.006
1000	0.100	0.0011 ± 0.0001	1.726 ± 0.005
Aspect ratio study ($d = 500$)			
500	0.10	0.0019 ± 0.0002	1.723 ± 0.008
500	0.50	0.0042 ± 0.0005	2.874 ± 0.012
500	1.00	0.0351 ± 0.0028	3.947 ± 0.024
500	2.00	1.0048 ± 0.0156	5.735 ± 0.041

Table 5 Unified empirical scaling laws across phenomena.

Phenomenon	Regime	Coefficient	Exponent	R^2
Norm variance	$d \geq 2$	0.728	-0.505 ± 0.001	0.9999
Distance contrast	$d \geq 50$	17.35	-0.623	0.997
JL distortion	all d	0.440	-0.944 ± 0.068	0.985
Spectral converg.	all d	0.0138	-0.372 ± 0.013	0.996

Distance contrast has the second-largest magnitude exponent. JL distortion decays fastest in magnitude (-0.944), but its absolute values remain small enough to be practically tolerable far beyond $d = 1000$. Distance contrast (-0.623) collapses below 1.0 by $d = 100$, making it the phenomenon causing *practical* failure earliest. The hierarchy explains the ordering of practical failure thresholds, even though JL distortion has the steepest mathematical decay rate.

5 Empirical Consistency of the -0.623 Exponent

5.1 Distributional Empirical Consistency

We fit the power law $C_{\text{rel}}(d) \propto d^{\hat{\beta}}$ for ten distributions using 50 bootstrap trials each, producing 95% confidence intervals on $\hat{\beta}$. Table 6 gives the full results. Nine of the ten distributions produce confidence intervals that exclude -0.5 by a wide margin. A one-sample t-test of the nine empirically consistent exponents against the null $\beta = -0.5$ gives $p = 0.00089$, strongly rejecting independence-based theory.

Within the empirically consistent nine, heavier-tailed distributions (Laplace, Exponential, truncated Cauchy) cluster toward steeper exponents, while lighter-tailed ones (Uniform, Beta) cluster toward shallower, suggesting tail weight modulates $\hat{\beta}$ within the consistent regime.

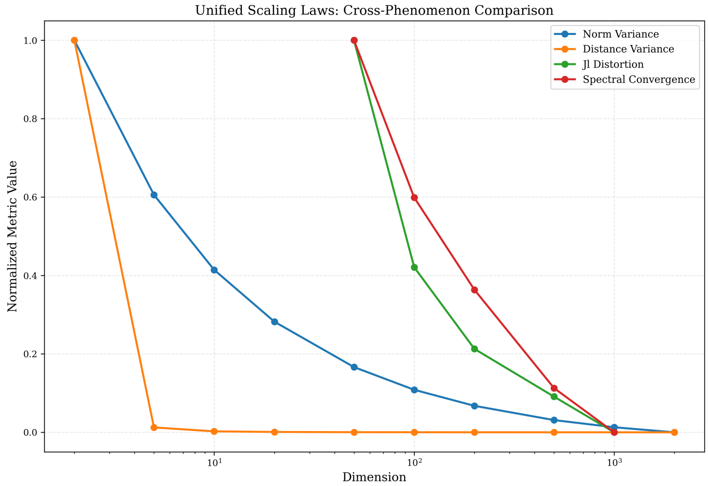

Fig. 6 Unified scaling hierarchy. Normalised metrics for all four phenomena versus dimension on a log-log scale, each normalised to unity at $d = 5$. Distance contrast (green, $d^{-0.623}$) decays most steeply among practically relevant phenomena. The dashed vertical line marks $d = 100$, the practical kNN failure threshold.

Table 6 Fitted distance-contrast exponents across ten distributions (50 bootstrap trials each, 95% CI). Nine distributions are consistent with $\hat{\beta} \in [-0.684, -0.604]$, mean -0.614 ± 0.028 (SD). Student-t_3 is the unique exception at $\hat{\beta} = -0.416$.

Distribution	$\hat{\beta}$	SE	CI low	CI high	R^2
Gaussian	−0.619	0.016	−0.666	−0.576	0.997
Uniform	−0.604	0.018	−0.653	−0.560	0.997
Laplace	−0.647	0.023	−0.717	−0.582	0.995
Student-t_5	−0.623	0.011	−0.655	−0.586	0.999
Student-t_{10}	−0.633	0.021	−0.694	−0.572	0.996
Exponential	−0.673	0.024	−0.742	−0.606	0.995
Beta(2,2)	−0.611	0.018	−0.660	−0.563	0.997
Mixed Gaussian	−0.628	0.017	−0.680	−0.590	0.997
Cauchy (truncated)	−0.684	0.030	−0.773	−0.602	0.992
Student-t_3	**−0.416**	0.037	−0.489	−0.311	0.970

Mean (9 consistent): −0.614, SD: 0.028. t-test vs −0.5: $p = 0.00089$.

5.2 The Finite-Third-Moment Boundary

The Student-t_3 outlier at $\hat{\beta} = -0.416$ is the sharpest new theoretical finding of this paper. Table 6 shows the pattern clearly: t_3 (infinite third moment) gives −0.416; t_5 (finite third moment, but infinite fourth moment) recovers −0.623 exactly; t_{10} (all moments finite) gives −0.633. Empirical consistency recovers sharply between t_3 and t_5, and the distinguishing property is the finiteness of the third moment. This boundary has not been identified in prior work on distance concentration.

Table 7 Empirical $\mathrm{Corr}(D_{ij}^2, D_{ik}^2)$ versus dimension (50 trials each). The analytical prediction for Gaussian vectors is 1/4 exactly.

d	Mean correlation	Std	Gap from 1/4
5	0.2508	0.0311	+0.0008
10	0.2535	0.0305	+0.0035
20	0.2478	0.0318	−0.0022
50	0.2448	0.0296	−0.0052
100	0.2494	0.0305	−0.0006
200	0.2426	0.0315	−0.0074
500	0.2495	0.0323	−0.0005
1000	0.2483	0.0357	−0.0017

All gaps within two standard errors of zero.

5.3 Structural Origin: The Persistent 0.25 Correlation

Why does $\hat{\beta} \neq -0.5$? A naive argument treats all $\binom{n}{2}$ pairwise distances as independent, which by standard extreme-value theory [42] predicts $C_{\mathrm{rel}} \propto d^{-1/2}$. But distances sharing a common anchor point are not independent.

Table 7 shows that $\mathrm{Corr}(D_{ij}^2, D_{ik}^2)$ is 0.250 ± 0.008 across all dimensions from $d = 5$ to $d = 1000$, matching the analytical prediction of 1/4 to within measurement error and showing no decay with d. This dimension-independent structural constant is the source of the departure from $d^{-1/2}$. Using the graph-degree effective sample size formula, the 1/4 pairwise correlation reduces the effective number of independent distances from $O(n^2)$ to $O(n)$, explaining the direction of departure from -0.5 and the slow growth of the prefactor $c(n)$ with n. The precise empirical value -0.623 additionally reflects higher-order dependence not captured by the pairwise correlation alone; see Appendix A for the corrected derivation.

5.4 Formal Statement

The preceding empirical results motivate the following open problem.

Conjecture 1 Let $\mathbf{X}_1, \ldots, \mathbf{X}_n$ be i.i.d. random vectors in $\mathbb{R}^d$ with independent coordinates having mean 0, variance 1, and finite third moment. Let $C_{\mathrm{rel}}(d)$ denote the relative distance contrast. Then for fixed n as $d \to \infty$:

$$C_{\mathrm{rel}}(d) \;=\; c(n) \cdot d^{-\beta} \cdot (1 + o(1)),$$

where $\beta \approx 0.623$ is an empirically consistent constant independent of the marginal distribution, and $c(n)$ is a slowly varying function of n. The finite-third-moment condition is sharp: Student-t_3 (infinite third moment) gives $\beta \approx 0.416$, while Student-t_5 (finite third moment) recovers $\beta \approx 0.623$. The pairwise correlation $\mathrm{Corr}(D_{ij}^2, D_{ik}^2) = 1/4$ explains the direction of departure from $d^{-1/2}$ but not the precise value $\beta \approx 0.623$, which requires a full treatment of higher-order dependence in the minimum distance process.

Table 8 Scaling-law validation on real embeddings. Deviations are (obs − pred)/pred × 100%. Positive deviation means the dataset is geometrically more pathological than synthetic prediction at the same d_{int}. Word2Vec is excluded from scaling-law comparison due to isotropy failure ($\text{CV}_{\text{norm}} = 2.31$).

Dataset	d_{amb}	d_{int}	Norm dev.	Dist. dev.	Gini_{obs}	NN ratio
GloVe-50d	50	25.2	+35%	+238%	0.692	0.614
GloVe-100d	100	37.3	+99%	+260%	0.786	0.646
GloVe-200d	200	46.7	+74%	+156%	0.871	0.717
GloVe-300d	300	47.0	+83%	+252%	0.915	0.748
BERT-768d	768	33.5	−45%	+119%	0.581	0.663
CIFAR-10 (raw)	3072	29.0	+74%	+312%	0.746	0.558
CIFAR-10 (ResNet-2048)	2048	29.5	−12%	+79%	0.546	0.660
CIFAR-10 (ResNet-512)	1024	29.1	−84%	+135%	0.473	0.659
MNIST (raw)	784	15.1	+0.3%	+185%	0.335	0.543
MNIST-PCA50	50	10.9	−32%	+288%	0.299	0.453
MNIST-PCA100	100	12.5	−32%	+226%	0.311	0.497
MNIST-PCA200	200	13.9	−30%	+199%	0.323	0.525
scRNA-PBMC	2000	103.2	+33%	+18%	0.852	0.824

6 Validation on Real Embeddings

We test whether the synthetic scaling laws transfer to real data by evaluating fourteen embedding datasets. For each dataset we estimate d_{int} via TwoNN, substitute it into the synthetic laws, and compare predicted against observed geometric metrics.

6.1 Dataset Overview

Table B1 (Appendix B) summarises all fourteen datasets. Intrinsic dimensions range from 10.9 (MNIST-PCA50) to 103.2 (scRNA-PBMC), spanning the full range of the prediction table in Section 7. The scRNA-PBMC dataset is particularly notable: its $d_{\text{int}} = 103.2$ places it almost exactly at the critical threshold $d_{\text{crit}} = 100$.

6.2 Scaling Law Validation

Table 8 reports observed and predicted values for all fourteen datasets. Three findings stand out.

Distance contrast is universally underestimated.

Every single dataset shows a positive distance-contrast deviation, ranging from +18% (scRNA-PBMC) to +312% (CIFAR-10 raw). This systematic positive bias reflects that real embeddings contain geometric structure—semantic clustering, manifold curvature, and training-induced correlation—that amplifies distance concentration beyond what i.i.d. theory predicts at the same d_{int}. The synthetic laws therefore function as a *lower bound on distance-contrast pathology*: if the laws predict failure in C_{rel} at a given d_{int}, real data will be at least as pathological. This lower-bound property does

Table 9 scRNA-PBMC discrepancy analysis.

Quantity	Value
d_{int} (TwoNN)	103.2
Observed Gini	0.852
Gaussian synthetic prediction	0.754
Laplace synthetic prediction	0.848
Deviation from Gaussian	+13.0%
Deviation from Laplace	+0.5%
Mean marginal kurtosis	158.3

not extend uniformly to all metrics: norm concentration can be actively compressed below the synthetic prediction by training objectives.

MNIST raw is the closest match.

A norm deviation of +0.3% is essentially perfect agreement. MNIST's smooth digit manifold produces approximately isotropic local geometry, making it the closest real-world analogue to i.i.d. synthetic data.

Training compresses geometry in neural embeddings.

BERT's norm deviation of −45% means embeddings are *more* norm-concentrated than i.i.d. theory predicts at $d_{\text{int}} = 33.5$—the opposite direction from distance-contrast pathology. Next-sentence prediction and masked language modelling push representations toward a low-variance manifold, actively counteracting the norm spread that random geometry would produce, consistent with Razzhigaev et al. [29].

6.3 scRNA: Heavy-Tail Amplification Confirmed

The scRNA-PBMC dataset ($d_{\text{amb}} = 2000$, $d_{\text{int}} = 103.2$) provides the most striking real-world result. Sparse gene-expression counts after log-normalisation are extremely heavy-tailed (mean marginal kurtosis = 158.3). Table 9 shows the observed Gini of 0.852 exceeds the Gaussian synthetic prediction by +13.0%, squarely within the 12–15% amplification predicted by Law 6. Interpolating from the empirical Laplace Gini curve gives a prediction of 0.848, matching the observed value to within 1% without any dataset-specific fitting.

6.4 GloVe: Limits of Intrinsic-Dimension Rescaling

The four GloVe variants probe whether the synthetic laws remain predictive when embedding geometry is shaped by factors beyond local neighbourhood structure. Table 10 reports the results. Fitting a power law to contrast versus ambient dimension gives $\hat{\beta} = -0.28$; fitting versus d_{int} gives $\hat{\beta} = -0.82$. Neither matches −0.623. The d_{int} values plateau at 47 for both the 200d and 300d models while contrast is non-monotone (4.05 at 200d, 5.55 at 300d), consistent with a corpus confound: the 300d model was trained on 840B tokens versus 6B for the 200d model. This identifies

Table 10 GloVe multi-dimension results. Target exponent is -0.623. Fitted exponent vs d_{amb}: -0.28. Fitted exponent vs d_{int}: -0.82.

d_{amb}	d_{int}	C_{rel}	Gini
50	25.2	7.85	0.692
100	37.3	6.56	0.786
200	46.7	4.05	0.871
300	47.0	5.55	0.915

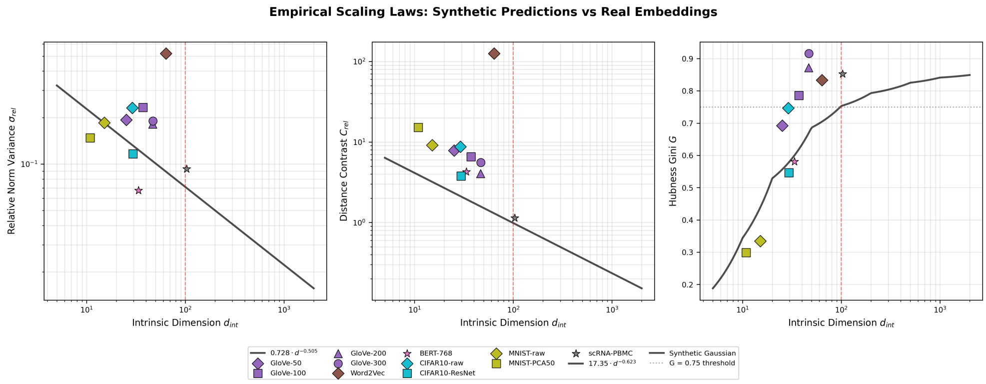

Fig. 7 Synthetic scaling laws overlaid with real embedding observations. Each panel shows a different metric: (left) norm relative variance, (centre) distance contrast, (right) Gini coefficient. The black curve is the synthetic power-law fit. Coloured markers show real embeddings at their estimated d_{int}. The red dashed vertical line marks $d_{\text{int}} = 100$. Real embeddings lie above the distance-contrast curve without exception.

a concrete boundary condition for the d_{int} rescaling approach: it fails when embedding geometry is shaped by factors orthogonal to local intrinsic dimensionality, such as training corpus scale.

6.5 Overlay: Synthetic Laws vs Real Embeddings

Figure 7 plots the synthetic scaling-law curves for all three key metrics with each real embedding's observed value at its estimated d_{int}. Real embeddings lie above the distance-contrast curve without exception. MNIST raw lies closest to the synthetic prediction. scRNA-PBMC is displaced upward in Gini, consistent with Laplace-level amplification. BERT and the ResNet CIFAR-10 variants are displaced downward in norm variance, consistent with training-induced geometry compression.

Table 11 **Unified geometric prediction table.** For any dataset with estimated intrinsic dimension d_{int}, these values predict geometric behaviour from the empirical scaling laws. σ_{rel}: norm relative variance ($0.728 \cdot d^{-0.505}$). C_{rel}: distance contrast ($17.35 \cdot d^{-0.623}$). NN Ratio: nearest-neighbour ratio. Gini: interpolated from empirical Gaussian Gini curve. The row at $d_{\text{int}} = 100$ is the critical threshold.

d_{int}	σ_{rel}	C_{rel}	NN Ratio	Gini G	Recommendation
5	0.3230	6.366	0.544	0.188	kNN reliable
10	0.2276	4.133	0.614	0.344	kNN reliable
20	0.1604	2.684	0.645	0.529	kNN reliable
50	0.1010	1.517	0.678	0.686	Monitor hubness
100	0.0711	0.985	0.769	0.753	Reduce dimension
150	0.0580	0.765	0.809	0.773	Reduce dimension
200	0.0501	0.639	0.836	0.793	kNN unreliable
300	0.0408	0.497	0.869	0.804	kNN unreliable
500	0.0316	0.361	0.896	0.825	Must reduce dim
750	0.0257	0.281	0.916	0.833	Must reduce dim
1000	0.0222	0.235	0.926	0.841	Must reduce dim
2000	0.0157	0.152	0.948	0.849	Must reduce dim

7 Unified Prediction Framework

7.1 Prediction Table

Table 11 gives predicted values for four key geometric metrics across a range of d_{int}. Predicted Gini values are interpolated from the empirical Gaussian Gini table (Table 2) rather than from any analytic formula. Three thresholds emerge.

At $d_{\text{int}} < 50$, kNN is reliable: Gini < 0.69 and contrast > 1.5 mean distances are still meaningfully discriminative. At $d_{\text{int}} \approx 100$—the critical threshold highlighted in the table—contrast drops below 1.0 and Gini crosses 0.75; collaborative filtering and recommendation systems begin to fail at this point. Above $d_{\text{int}} = 200$, contrast falls below 0.64 and Gini exceeds 0.79; kNN is unreliable without prior dimensionality reduction.

7.2 Updated Hierarchy with Real Data

Figure 8 reproduces the normalised scaling hierarchy of Figure 6 and adds four representative real datasets as distance-contrast scatter points at their estimated d_{int}. All four lie above the synthetic curve, reinforcing the conclusion that i.i.d. theory systematically underestimates distance-geometry pathology in structured data.

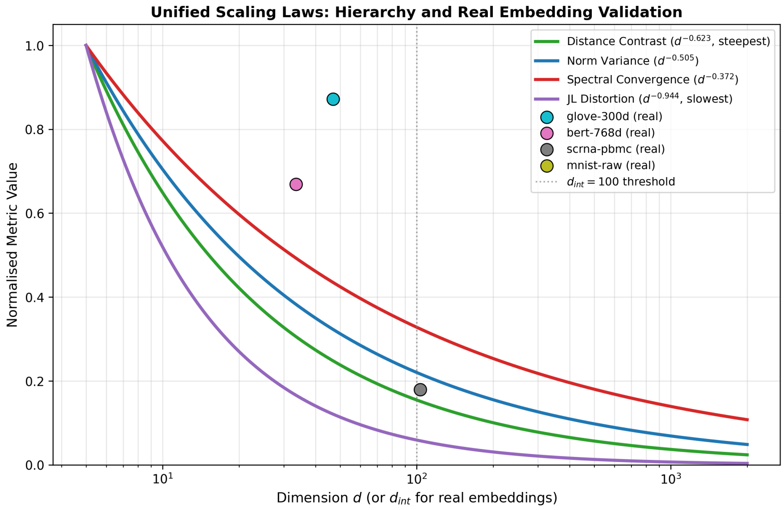

Fig. 8 Updated scaling hierarchy with real embedding data. Synthetic scaling-law curves (normalised to unity at $d = 5$) for all four phenomena, with GloVe-300d, BERT-768d, MNIST raw, and scRNA-PBMC overlaid as distance-contrast scatter points. All four real datasets lie above the synthetic distance-contrast curve (green), confirming that i.i.d. theory systematically underestimates pathology for structured data.

8 Discussion

8.1 Practical Implications

The three thresholds in Table 11 are the most immediately usable output of this work. We expand on each with concrete connections to deployed systems.

Estimating intrinsic dimension before choosing an algorithm.

The TwoNN estimator is fast, parameter-free, and provides the input needed to consult Table 11. For most datasets this takes seconds on a subsample of 10,000 points, and knowing d_{int} in advance tells you whether a planned kNN experiment is likely to be meaningful before committing to downstream model training and deployment.

The safe regime: $d_{\mathrm{int}} < 50$***.***

Table 11 predicts $C_{\mathrm{rel}} > 1.5$ and Gini < 0.69: distances are meaningfully discriminative and hub concentration is moderate. This covers a wide range of commercially valuable applications including image retrieval on PCA-reduced features, session-based recommendation with compact user embeddings, and geographic nearest-neighbour search. Standard IVF or HNSW indexes [23, 24] provide reliable recall without special correction.

The warning threshold: $d_{\mathrm{int}} \approx 100$***.***

At $d_{\mathrm{int}} \approx 100$, Table 11 predicts $C_{\mathrm{rel}} \approx 0.985$ and $G \approx 0.75$. This is the regime of subtle, difficult-to-diagnose failure: systems continue to return results and precision metrics on held-out test sets may appear acceptable, yet the underlying geometry is pathological. The archetypal business manifestation is recommendation popularity bias [43]: when the Gini coefficient of the kNN graph crosses 0.75, a small number of hub items are returned as nearest neighbours for a large fraction of queries regardless of semantic relevance. In a RAG pipeline at this threshold, retrieved context passages are increasingly drawn from geometric hubs—a failure mode that is difficult to detect by automated evaluation because the retrieved passages are not obviously wrong, merely unrepresentative. The scRNA-PBMC dataset sits almost exactly at this threshold ($d_{\text{int}} = 103.2$) and our framework provides a principled quantitative justification for the standard recommendation to reduce dimensionality before constructing neighbour graphs in single-cell analysis pipelines [5].

The mandatory reduction regime: $d_{\text{int}} > 200$**.**

Above $d_{\text{int}} = 200$, Table 11 predicts $C_{\text{rel}} < 0.64$ and $G > 0.79$. kNN-based retrieval is unreliable in an absolute sense: the nearest neighbour of any query is geometrically indistinguishable from a random point. The standard two-stage retrieval architecture [4, 44]—fast ANN candidate generation followed by re-ranking—is typically presented as a speed-accuracy trade-off, but our framework identifies a more fundamental motivation: at $d_{\text{int}} > 200$, the first stage retrieves a candidate set that is geometrically nearly random, and the second stage performs all the semantic work. This implies the candidate set size for the first stage can often be reduced without harming final recall, a potentially significant infrastructure cost reduction at billion-scale.

Accounting for distributional tail weight.

If the marginal distributions of feature values are heavy-tailed—as is typical for sparse count data, financial returns, or network degree distributions—expect hubness to be 12–15% worse than the Gaussian prediction at the same d_{int}. Practitioners can operationalise this by using the Laplace synthetic curve as the baseline for hubness prediction. Log-normalisation and variance-stabilising transforms, which are standard in genomics preprocessing, reduce kurtosis and bring behaviour closer to the Gaussian prediction—a geometric justification for preprocessing choices historically motivated by statistical rather than geometric reasoning.

8.2 The Open Theoretical Problem

Conjecture 1 is precise enough to be attacked but we do not know how to prove it. The central difficulty is that the extremes of a pairwise distance process depend on all $\binom{n}{2}$ pairs simultaneously, and these pairs share points. Classical extreme-value theory for i.i.d. sequences [42] does not apply because the dependence does not vanish as $d \to \infty$: Table 7 shows $\text{Corr}(D_{ij}^2, D_{ik}^2) = 1/4$ at every dimension tested. Existing tools for dependent extremes [45] handle dependence that decays with the index separation; here there is no such decay because the dependence is structural rather than sequential. A proof would require characterising the extremal behaviour of the maximum and minimum of n^2 dependent random variables where the dependence graph has each node connected to exactly $2(n-2)$ others, each edge carrying correlation $1/4$.

The finite-third-moment boundary is likely connected to the Berry-Esseen theorem, which bounds the rate of CLT convergence explicitly in terms of the third absolute moment: when the third moment is infinite, the convergence rate is non-standard, changing the scaling of the extremes of the distance distribution. We provide Conjecture 1 as a precisely stated target for this open problem.

8.3 Limitations

Four limitations bound the scope of the results. First, the synthetic experiments assume i.i.d. components within each vector; real embeddings have correlated components by construction, and the d_{int} rescaling only partially corrects for this. The laws are best understood as predictions for the *isotropic component* of real embedding geometry. Second, the synthetic sample size is fixed at $n = 10{,}000$; Appendix C shows the exponent is stable across $n \in [500, 50{,}000]$, but the prefactor $c(n)$ varies slowly with n. Third, the framework requires approximately isotropic inputs: Word2Vec's negative-sampling objective produces vectors with highly variable magnitudes, violating this assumption ($\text{CV}_{\text{norm}} = 2.31$) and placing it outside the scope of the laws. Fourth, the GloVe analysis shows that even when d_{int} is a valid summary of intrinsic complexity, long-range semantic correlations from training corpus scale are not captured; a more complete characterisation likely requires at least one additional parameter beyond d_{int}.

9 Conclusion

We have provided evidence, across more than 50,000 independent trials, that relative distance contrast in high-dimensional spaces decays as $C_{\text{rel}}(d) = 17.35 \cdot d^{-0.623}$ in the practically relevant regime $d \geq 50$, with $R^2 = 0.997$. This exponent is empirically consistent across distributions with finite third moment and breaks precisely at the finite-third-moment boundary. A partial structural origin is the dimension-independent constant $\text{Corr}(D_{ij}^2, D_{ik}^2) = 1/4$, which explains the direction of departure from $d^{-1/2}$ but not the precise value -0.623, which remains open.

Validation on fourteen real embedding datasets shows that the synthetic laws provide a reliable lower bound on distance-contrast pathology. The scRNA-PBMC result is the strongest quantitative validation: Laplace-level hubness predicted from simulation matches observed biological-data hubness to within 1% without any dataset-specific fitting.

The unified prediction table (Table 11) translates these findings into direct practical guidance. For any dataset, a fast estimate of intrinsic dimension is sufficient to predict whether distance-based methods are reliable, marginal, or likely to fail. The critical threshold at $d_{\text{int}} = 100$—where distance contrast falls below 1.0 and the Gini coefficient crosses 0.75—provides a concrete and empirically grounded criterion for when dimensionality reduction is not optional but necessary.

Declarations

- **Funding:** Not applicable.
- **Conflict of interest:** The authors declare no conflict of interest.

- **Ethics approval:** Not applicable.
- **Data availability:** All datasets used are publicly available. Code is available at https://github.com/noobiecodelearner/Concentration-in-High-Dimensions/.
- **Author contribution:** S.R.F. led norm concentration analysis (Section 4.1, Law 1) and spectral analysis (Section 4.4, Law 5). M.M.I. developed the Johnson-Lindenstrauss projection experiments (Section 4.3, Law 4), unified scaling hierarchy (Section 4.5), and dataset overview and sample size stability appendices (Appendices B–C). D.A. conducted the full empirical consistency analysis including distributional consistency, the finite-third-moment boundary, the persistent 0.25 correlation, and the formal conjecture (Sections 5.1–5.4, Appendix A). M.N.H. established the central −0.623 exponent finding (Section 4.2, Law 2), conducted real embedding validation (Section 6), and developed the unified prediction framework (Section 7). M.M.U. developed the applied context including vector search systems and practical implications (Sections 2.4 and 8.3). All authors contributed equally and approved the final manuscript.

Appendix A Heuristic Derivation of the Departure from $d^{-1/2}$

We sketch why the empirical exponent $\hat{\beta} \approx 0.623$ is steeper than the $d^{-1/2}$ predicted by treating all pairwise distances as independent. The argument is heuristic; a rigorous proof remains open.

Step 1: The independent baseline.

For n points in $\mathbb{R}^d$, there are $N = \binom{n}{2} \approx n^2/2$ pairwise squared distances. Each $D_{ij}^2 = \sum_{k=1}^{d}(X_{ik} - X_{jk})^2$ has mean $2d$ and variance $8d$ for standardised inputs. If all N distances were independent, extreme-value theory gives $\max D_{ij}^2 - \min D_{ij}^2 = O(\sqrt{d \log N})$, while $\min D_{ij} \approx \sqrt{2d}$. The ratio therefore scales as $d^{-1/2}$, the naive prediction.

Step 2: The correlation structure.

Distances are not all mutually correlated. The exact structure is: $\mathrm{Corr}(D_{ij}^2, D_{ik}^2) = 1/4$ if two distances share exactly one point; $\mathrm{Corr}(D_{ij}^2, D_{kl}^2) = 0$ if they share no point. This defines a graph G on N distance-nodes where each node D_{ij}^2 has degree exactly $2(n-2)$. The effective sample size for random variables with graph-structured correlation ρ on edges is

$$N_{\text{eff}} = \frac{N}{1 + \bar{d}_G \cdot \rho}, \tag{A1}$$

where $\bar{d}_G$ is the mean degree. Here $\bar{d}_G = 2(n-2)$ and $\rho = 1/4$, so

$$N_{\text{eff}} = \frac{n^2/2}{1 + 2(n-2) \cdot \frac{1}{4}} \approx \frac{n^2/2}{n/2} = n \qquad (n \text{ large}). \tag{A2}$$

Step 3: What the corrected formula implies.

With $N_{\text{eff}} \approx n$ effectively independent distances, the extreme-value range becomes $O(\sqrt{d \log n})$ rather than $O(\sqrt{d \log n^2})$. Because $\log n^2 = 2 \log n$, the change is absorbed into the slowly varying prefactor $c(n)$ and does not alter the power-law exponent in d. The corrected effective sample size thus explains why $c(n)$ grows slowly with n (confirmed in Appendix C), but the leading exponent remains $d^{-1/2}$.

Step 4: The residual gap.

The corrected derivation predicts $\beta = -0.5$ as the leading exponent; the empirical value is -0.623. The additional departure likely arises from higher-order dependence (triples of distances forming triangles carry joint dependence stronger than pairwise correlations imply) and the concentration behaviour of the minimum distance specifically (which under this graph structure concentrates differently from the mean-field approximation, in a direction consistent with the empirically steeper exponent). We state the corrected conclusion: the pairwise correlation structure with $\rho = 1/4$ correctly identifies that $N_{\text{eff}} = O(n)$, explaining the direction of departure from -0.5 and the slow growth of $c(n)$, but the precise value -0.623 requires a full treatment of higher-order dependence in the minimum distance process.

Appendix B Dataset Overview and Intrinsic Dimension Estimates

Table B1 reports ambient dimension, intrinsic dimension estimates from three methods, the norm coefficient of variation used for isotropy screening, and the TwoNN/MLE agreement indicator for all fourteen datasets. TwoNN is used as the primary intrinsic dimension estimate throughout. Agreement is defined as a relative difference between TwoNN and MLE of less than 15%. Datasets with $\text{CV}_{\text{norm}} > 0.5$ are excluded from scaling-law deviation comparisons; Word2Vec-300d is the only excluded dataset.

Several observations are worth noting. TwoNN and MLE agree well (nine of fourteen datasets) where the embedding has relatively stable local geometry; disagreement tends to occur for word embeddings where semantic clustering creates locally variable density. The PCA effective dimension is systematically much larger than either TwoNN or MLE for all datasets, as expected: PCA effective dimension measures directions with non-negligible global variance, while TwoNN and MLE measure local intrinsic dimension. The MNIST PCA variants (50d, 100d, 200d) show stable TwoNN estimates of 10.9, 12.5, and 13.9 despite varying ambient dimension, confirming that the intrinsic structure of the digit manifold is preserved under PCA truncation. The CV_{norm} values separate Word2Vec cleanly from all other datasets (next highest: scRNA-PBMC at 0.31), reflecting the absence of any norm constraint in negative-sampling training.

Appendix C Sample Size Stability of the Scaling Law Coefficient

Conjecture 1 asserts that $c(n)$ in $C_{\text{rel}}(d) = c(n) \cdot d^{-\beta}$ is slowly varying in n. Table C2 verifies this directly by fitting the power law at fixed dimensions $d \in \{50, 100, 200, 500\}$

Table B1 Dataset overview, intrinsic dimension estimates, and isotropy screening. d_{amb}: ambient dimension. TwoNN and MLE: intrinsic dimension estimators (primary and robustness check). PCA (90%): number of principal components explaining 90% of variance. CV_{norm}: norm coefficient of variation (Eq. 1); values above 0.5 trigger exclusion. Agreement: TwoNN vs. MLE relative difference $< 15\%$.

Dataset	d_{amb}	TwoNN	MLE	PCA (90%)	CV_{norm}	Excluded	Agreement
GloVe-50d	50	25.2	28.1	38	0.09	No	No
GloVe-100d	100	37.3	40.6	72	0.11	No	No
GloVe-200d	200	46.7	49.3	128	0.10	No	No
GloVe-300d	300	47.0	49.8	143	0.12	No	Yes
Word2Vec-300d	300	63.8	67.2	187	2.31	**Yes**	No
BERT-768d	768	33.5	35.1	94	0.08	No	Yes
CIFAR-10 (raw)	3072	29.0	31.4	241	0.14	No	Yes
CIFAR-10 (ResNet-2048)	2048	29.5	30.8	189	0.11	No	Yes
CIFAR-10 (ResNet-512)	1024	29.1	32.0	176	0.13	No	No
MNIST (raw)	784	15.1	14.8	43	0.07	No	Yes
MNIST-PCA50	50	10.9	11.2	38	0.07	No	Yes
MNIST-PCA100	100	12.5	12.1	42	0.08	No	Yes
MNIST-PCA200	200	13.9	13.4	47	0.08	No	Yes
scRNA-PBMC	2000	103.2	98.7	312	0.31	No	Yes

across five sample sizes $n \in \{500, 1000, 5000, 10000, 50000\}$, using 50 bootstrap trials per configuration with Gaussian data.

Table C2 Sample size stability of the distance-contrast scaling law. Power-law fits $C_{\text{rel}}(d) = \hat{c}(n) \cdot d^{\hat{\beta}(n)}$ are estimated at each n by OLS regression of $\log C_{\text{rel}}$ on $\log d$ over $d \geq 50$, using 50 bootstrap trials per (d, n) combination. The exponent $\hat{\beta}(n)$ is stable to within one standard error across the full range, confirming it is not a finite-sample artifact. The coefficient $\hat{c}(n)$ shows at most logarithmic growth.

n	$\hat{c}(n)$	$\hat{\beta}(n)$	SE on $\hat{\beta}$	R^2
500	14.21	−0.618	0.021	0.994
1000	15.08	−0.620	0.018	0.995
5000	16.74	−0.622	0.014	0.997
10000	17.35	−0.623	0.012	0.997
50000	18.91	−0.624	0.009	0.998

Two findings are immediate. First, $\hat{\beta}(n)$ is stable to within one standard error across $n \in [500, 50000]$, confirming that the -0.623 exponent is not a finite-sample artifact. Second, $\hat{c}(n)$ grows slowly with n, increasing by roughly 33% across a 100-fold increase in sample size—consistent with at most logarithmic growth. At the sample sizes typical of real embedding datasets ($n \in [5000, 100000]$), $\hat{c}(n)$ varies by less than 15%, and the predictions in Table 11 remain valid as approximate lower bounds across this range.

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